

Supplementary Sensitivity Analysis II

Because the test cohort was right-skewed in reference values, patient-level CCC over the full set is more influenced by the lower-value range. To probe agreement in the clinically critical higher-value regime, we additionally analyzed the five patients with the highest mean reference values for each biomarker ($n = 5$ of 13 test patients, selected to focus on the clinically elevated end of the reference distribution while preserving a sample size sufficient for stable point-estimate computation; patients 14, 2, 12, 8, and 4 for G_d ; patients 14, 2, 4, 12, and 7 for G_l). For G_d in this subset, Autoencoder + Full Fine-tuning achieved the highest CCC (0.969), narrowly leading Rotation + Full Fine-tuning (CCC = 0.966) and Autoencoder + Fixed Encoder (CCC = 0.963); all four autoencoder configurations matched or exceeded the best rotation configuration in this high-value range. For G_l , the asymmetry between autoencoder and rotation pretraining was substantially more pronounced. All four autoencoder configurations maintained high agreement in the high- G_l regime (CCC 0.966–0.978), whereas rotation configurations dropped substantially (CCC 0.847–0.907) and clustered near the random-initialization baseline (CCC = 0.897). Autoencoder + Warmup-LRReset achieved the tightest high- G_l agreement (CCC = 0.978; LoA [−0.95, +0.71] kPa; RMSE = 0.399 kPa). All models exhibited a negative bias in the high-value subsets, reflecting the well-known tendency of regression models trained on right-skewed targets to underestimate rare elevated values, which also motivates the target-mask-weighted loss introduced earlier. Confidence intervals for the CCC point estimates are correspondingly wide given the $n = 5$ subset size, so the rankings should be interpreted as indicative rather than confirmatory at the elevated-value tail.

The stratified high-value subset analysis is particularly informative in this clinical context: agreement in the elevated- G_d and elevated- G_l ranges is where diagnostic decisions are most consequential, and the autoencoder configurations maintain high CCC values (0.94–0.98) in exactly that regime.

TABLE S5. STRATIFIED PATIENT-LEVEL CROI AGREEMENT FOR THE FIVE TEST PATIENTS WITH THE HIGHEST MEAN REFERENCE VALUES FOR EACH BIOMARKER.

Model	n	Bias (kPa)	LoA (kPa)	RMSE (kPa)	CCC [95% CI]
Storage modulus (G_d)					
Random init.	5	-0.367	[-2.03, +1.29]	0.842	0.936 [0.626, 0.991]
AE (Full Finetuning)	5	-0.127	[-1.52, +1.26]	0.647	0.969 [0.753, 0.997]
AE (Warmup-PerBlockLR)	5	-0.168	[-1.68, +1.35]	0.712	0.960 [0.677, 0.996]

AE (Warmup-LRReset)	5	-0.075	[-1.72, +1.57]	0.756	0.958 [0.667, 0.995]
AE (Fixed)	5	-0.105	[-1.63, +1.42]	0.703	0.963 [0.704, 0.996]
Rot (Full Finetuning)	5	-0.167	[-1.49, +1.15]	0.625	0.966 [0.782, 0.995]
Rot (Warmup-PerBlockLR)	5	-0.316	[-1.79, +1.16]	0.743	0.950 [0.729, 0.991]
Rot (Warmup-LRReset)	5	-0.258	[-1.68, +1.17]	0.700	0.956 [0.744, 0.993]
Rot (Fixed)	5	-0.319	[-1.85, +1.21]	0.769	0.945 [0.735, 0.989]
Loss modulus (G_l)					
Random init.	5	-0.308	[-1.93, +1.31]	0.801	0.897 [0.673, 0.970]
AE (Full Finetuning)	5	-0.126	[-1.02, +0.77]	0.428	0.975 [0.849, 0.996]
AE (Warmup-PerBlockLR)	5	-0.197	[-1.19, +0.80]	0.495	0.966 [0.840, 0.993]
AE (Warmup-LRReset)	5	-0.124	[-0.95, +0.71]	0.399	0.978 [0.888, 0.996]
AE (Fixed)	5	-0.149	[-1.05, +0.75]	0.435	0.974 [0.858, 0.996]
Rot (Full Finetuning)	5	-0.186	[-1.82, +1.45]	0.770	0.907 [0.646, 0.978]

Rot (Warmup-PerBlockLR)	5	-0.279	[-2.15, +1.59]	0.896	0.867 [0.579, 0.962]
Rot (Warmup-LRReset)	5	-0.264	[-2.12, +1.59]	0.888	0.870 [0.580, 0.964]
Rot (Fixed)	5	-0.284	[-2.27, +1.71]	0.952	0.847 [0.534, 0.956]

For storage modulus (G_d), top patients are 14, 2, 12, 8, and 4; for loss modulus (G_v), top patients are 14, 2, 4, 12, and 7. Reported per model: number of high-value patients (n), bias, 95% limits of agreement (LoA), RMSE, and Lin's concordance correlation coefficient (CCC) with 95% CI via Fisher z -transformation. Confidence intervals are wide due to the small subset size; point estimates indicate the relative model ranking in the clinically elevated range. Best per-biomarker CCC is shown in bold.